EduX

School Management System

Database Schema

Developer Documentation

Version 1.0
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Chapter 1

Database Overview

1.1 Introduction

EduX uses SQLite as its database engine, accessed through the Drift ORM (formerly Moor). The database is stored locally on the user's machine, enabling offline-first functionality.

1.2 Database Statistics

Metric	Value
Total Tables	27+
Database Engine	SQLite 3
ORM	Drift (dart package)
Location	`AppData/Roaming/EduX/edux_database.sqlite`

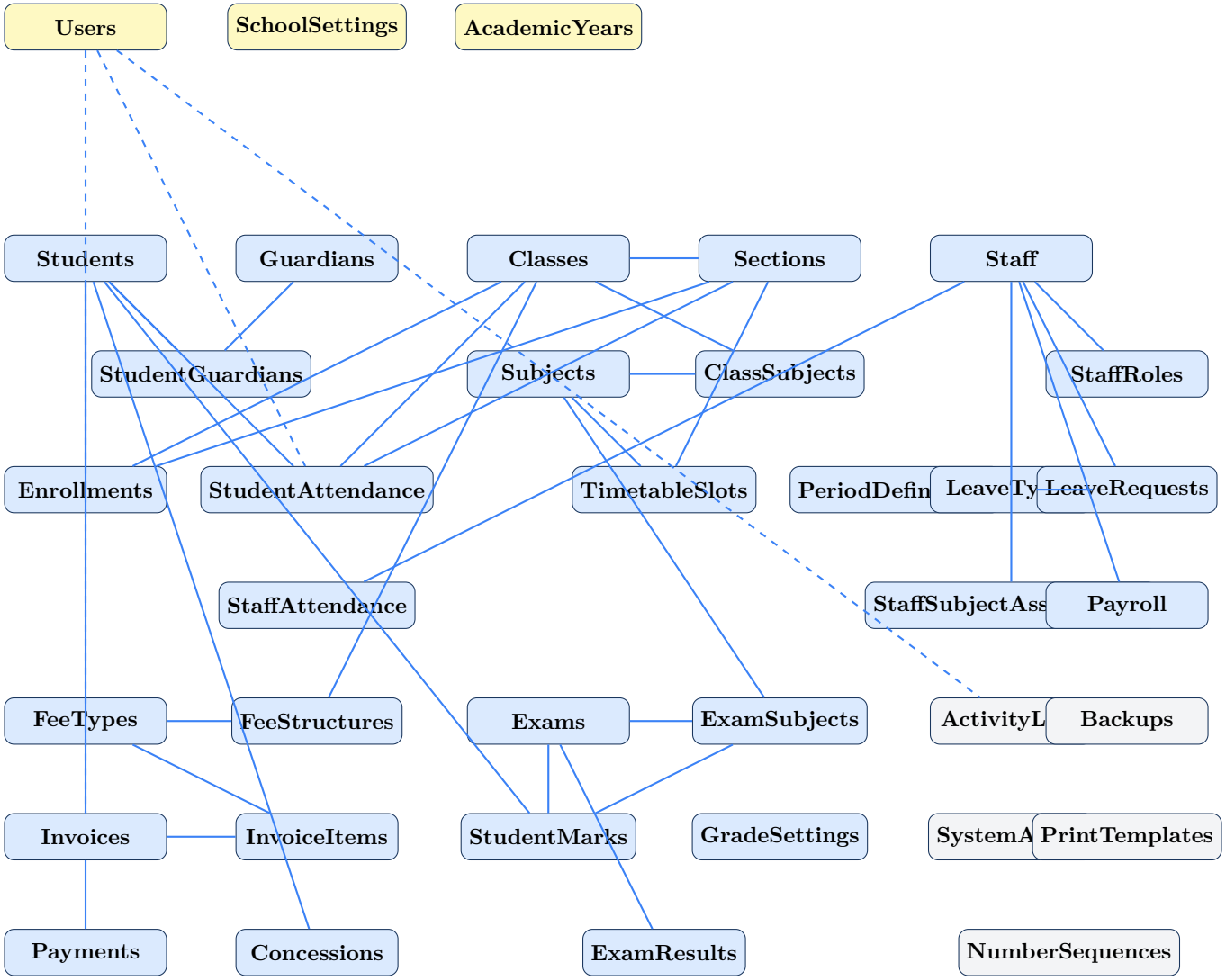
1.3 Table Categories

Core 3 tables	**Student** 4 tables	**Academic** 6 tables	**Attendance** 2 tables
Fee 6 tables	**Exam** 5 tables	**Staff** 7 tables	**System** 5 tables

Chapter 2

Entity Relationship Diagram

2.1 Complete ER Diagram


Users
SchoolSettings
AcademicYears
Students
Guardians
Classes
Sections
Staff
StudentGuardians
Subjects
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StaffRoles
Enrollments
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GradeSettings
SystemA
PrintTemplates
Payments
Concessions
ExamResults
NumberSequences

Chapter 3

Core Tables

3.1 SchoolSettings

Stores school configuration and settings.

Column	Type	Description
id	INTEGER	Primary key (auto-increment)
schoolName	TEXT	School name (max 200 chars)
logo	BLOB	School logo image
address	TEXT	School address
city	TEXT	City
state	TEXT	State/Province
postalCode	TEXT	Postal code
country	TEXT	Country (default: Pakistan)
phone	TEXT	Primary phone
email	TEXT	Email address
principalName	TEXT	Principal's name
currencySymbol	TEXT	Currency (default: PKR)
currentAcademicYear	TEXT	Current year (e.g., 2025-2026)
workingDays	TEXT	Comma-separated days
schoolStartTime	TEXT	HH:mm format
schoolEndTime	TEXT	HH:mm format
isSetupComplete	BOOLEAN	Setup wizard completed
createdAt	DATETIME	Record creation time
updatedAt	DATETIME	Last update time

3.2 Users

User accounts with authentication.

Column	Type	Description
id	INTEGER	Primary key
uuid	TEXT	Unique external identifier
username	TEXT	Login username (unique)
passwordHash	TEXT	SHA-256 password hash
passwordSalt	TEXT	Password salt

Column	Type	Description
fullName	TEXT	Display name
email	TEXT	Email address
phone	TEXT	Phone number
role	TEXT	admin/principal/teacher/accountant
photo	BLOB	Profile photo
isActive	BOOLEAN	Account active status
isSystemAdmin	BOOLEAN	Cannot be deleted
lastLogin	DATETIME	Last login timestamp
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

3.3 AcademicYears

Academic year configurations.

Column	Type	Description
id	INTEGER	Primary key
name	TEXT	Year name (e.g., 2025-2026)
startDate	DATETIME	Year start date
endDate	DATETIME	Year end date
isCurrent	BOOLEAN	Currently active year
isArchived	BOOLEAN	Archived status
createdAt	DATETIME	Creation time

Chapter 4

Student Tables

4.1 Students

Complete student information.

Column	Type	Description
id	INTEGER	Primary key
uuid	TEXT	Unique identifier
admissionNumber	TEXT	Unique admission number
firstName	TEXT	First name (1-50 chars)
lastName	TEXT	Last name (1-50 chars)
dateOfBirth	DATETIME	Birth date
gender	TEXT	male/female
bloodGroup	TEXT	Blood group
religion	TEXT	Religion
nationality	TEXT	Nationality (default: Pakistani)
cnic	TEXT	CNIC/B-Form number
address	TEXT	Residential address
city	TEXT	City
phone	TEXT	Contact phone
email	TEXT	Email address
photo	BLOB	Student photo
medicalInfo	TEXT	Medical notes
allergies	TEXT	Known allergies
specialNeeds	TEXT	Special requirements
previousSchool	TEXT	Previous school name
status	TEXT	active/inactive/graduated/withdrawn
admissionDate	DATETIME	Admission date
leavingDate	DATETIME	Leaving date (if applicable)
leavingReason	TEXT	Reason for leaving
notes	TEXT	Additional notes
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

4.2 Guardians

Parent/Guardian information.

Column	Type	Description
id	INTEGER	Primary key
uuid	TEXT	Unique identifier
firstName	TEXT	First name
lastName	TEXT	Last name
relation	TEXT	father/mother/guardian/other
cnic	TEXT	CNIC number
phone	TEXT	Primary phone (required)
alternatePhone	TEXT	Secondary phone
email	TEXT	Email address
occupation	TEXT	Job/Profession
workplace	TEXT	Employer name
address	TEXT	Address
city	TEXT	City
photo	BLOB	Guardian photo
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

4.3 StudentGuardians

Many-to-many relationship between students and guardians.

Column	Type	Description
studentId	INTEGER	FK to Students (cascade delete)
guardianId	INTEGER	FK to Guardians (cascade delete)
isPrimary	BOOLEAN	Primary guardian flag
canPickup	BOOLEAN	Can pickup student
isEmergencyContact	BOOLEAN	Emergency contact flag
createdAt	DATETIME	Creation time
Primary Key: (studentId, guardianId)		

4.4 Enrollments

Student class enrollment history.

Column	Type	Description
id	INTEGER	Primary key
studentId	INTEGER	FK to Students
classId	INTEGER	FK to Classes
sectionId	INTEGER	FK to Sections
academicYear	TEXT	Academic year
rollNumber	TEXT	Roll number in class

Column	Type	Description
enrollmentDate	DATETIME	Enrollment date
endDate	DATETIME	End date (if applicable)
status	TEXT	active/promoted/transferred/withdrawn
isCurrent	BOOLEAN	Current enrollment flag
notes	TEXT	Notes about enrollment
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

Chapter 5

Academic Tables

5.1 Classes

School class definitions.

Column	Type	Description
id	INTEGER	Primary key
name	TEXT	Class name (e.g., Class 5)
level	TEXT	pre_primary/primary/middle/secondary
gradeLevel	INTEGER	Numeric level for sorting
displayOrder	INTEGER	Display order
description	TEXT	Class description
monthlyFee	REAL	Default monthly fee
isActive	BOOLEAN	Active status
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

5.2 Sections

Class sections (A, B, C, etc.).

Column	Type	Description
id	INTEGER	Primary key
classId	INTEGER	FK to Classes
name	TEXT	Section name (e.g., A)
capacity	INTEGER	Maximum students
classTeacherId	INTEGER	FK to Staff (optional)
roomNumber	TEXT	Room number/name
isActive	BOOLEAN	Active status
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

5.3 Subjects

Subject definitions.

Column	Type	Description
id	INTEGER	Primary key
code	TEXT	Subject code (e.g., ENG)
name	TEXT	Subject name
type	TEXT	core/elective/optional
creditHours	INTEGER	Credit hours per week
description	TEXT	Description
isActive	BOOLEAN	Active status
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

Chapter 6

Fee Tables

6.1 FeeTypes

Fee category definitions.

Column	Type	Description
id	INTEGER	Primary key
name	TEXT	Fee type name
description	TEXT	Description
isMonthly	BOOLEAN	Monthly recurring fee
isRefundable	BOOLEAN	Refundable fee
accountCode	TEXT	Accounting code
isActive	BOOLEAN	Active status
displayOrder	INTEGER	Display order
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

6.2 Invoices

Fee invoices.

Column	Type	Description
id	INTEGER	Primary key
invoiceNumber	TEXT	Unique invoice number
studentId	INTEGER	FK to Students
month	TEXT	Invoice month (YYYY-MM)
academicYear	TEXT	Academic year
totalAmount	REAL	Sum of all fee items
discountAmount	REAL	Discount from concessions
netAmount	REAL	Total - discount
paidAmount	REAL	Amount paid so far
balanceAmount	REAL	Net - paid
status	TEXT	pending/partial/paid/overdue
issueDate	DATETIME	Invoice issue date
dueDate	DATETIME	Payment due date
lastPaymentDate	DATETIME	Last payment date

Column	Type	Description
notes	TEXT	Notes
generatedBy	INTEGER	FK to Users
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

6.3 Payments

Payment transactions.

Column	Type	Description
id	INTEGER	Primary key
receiptNumber	TEXT	Unique receipt number
invoiceId	INTEGER	FK to Invoices
studentId	INTEGER	FK to Students
amount	REAL	Payment amount
paymentMode	TEXT	cash/bank_transfer/cheque/online
referenceNumber	TEXT	Transaction reference
bankName	TEXT	Bank name
chequeDate	DATETIME	Cheque date
remarks	TEXT	Payment remarks
paymentDate	DATETIME	Payment date
receivedBy	INTEGER	FK to Users
isCancelled	BOOLEAN	Cancelled flag
cancellationReason	TEXT	Cancellation reason
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

Chapter 7

Examination Tables

7.1 Exams

Exam definitions.

Column	Type	Description
id	INTEGER	Primary key
uuid	TEXT	Unique identifier
name	TEXT	Exam name
type	TEXT	unit_test/term_exam/annual_exam
academicYear	TEXT	Academic year
classId	INTEGER	FK to Classes
startDate	DATETIME	Exam start date
endDate	DATETIME	Exam end date
description	TEXT	Description
status	TEXT	draft/active/completed
createdBy	INTEGER	FK to Users
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

7.2 StudentMarks

Marks obtained by students.

Column	Type	Description
id	INTEGER	Primary key
examId	INTEGER	FK to Exams
examSubjectId	INTEGER	FK to ExamSubjects
studentId	INTEGER	FK to Students
marksObtained	REAL	Marks obtained
isAbsent	BOOLEAN	Absent flag
grade	TEXT	Calculated grade
remarks	TEXT	Remarks
enteredBy	INTEGER	FK to Users
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

Column	Type	Description
Unique: (examSubjectId, studentId)		

7.3 GradeSettings

Grading criteria.

Column	Type	Description
id	INTEGER	Primary key
grade	TEXT	Grade name (e.g., A+)
minPercentage	REAL	Minimum percentage
maxPercentage	REAL	Maximum percentage
gpa	REAL	GPA points
remarks	TEXT	Report card remarks
displayOrder	INTEGER	Display order
isPassing	BOOLEAN	Passing grade flag
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

Chapter 8

Staff Tables

8.1 Staff

Employee information.

Column	Type	Description
id	INTEGER	Primary key
uuid	TEXT	Unique identifier
employeeId	TEXT	Unique employee ID
firstName	TEXT	First name
lastName	TEXT	Last name
dateOfBirth	DATETIME	Birth date
gender	TEXT	male/female
cnic	TEXT	CNIC number
phone	TEXT	Primary phone
email	TEXT	Email address
address	TEXT	Address
qualification	TEXT	Highest qualification
specialization	TEXT	Subject expertise
designation	TEXT	Job title
department	TEXT	Department
roleId	INTEGER	FK to StaffRoles
basicSalary	REAL	Monthly basic salary
joiningDate	DATETIME	Joining date
status	TEXT	active/on_leave/resigned
userId	INTEGER	FK to Users (optional)
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time

8.2 Payroll

Salary records.

Column	Type	Description
id	INTEGER	Primary key
staffId	INTEGER	FK to Staff

Column	Type	Description
month	TEXT	Payroll month (YYYY-MM)
basicSalary	REAL	Basic salary
allowances	REAL	Total allowances
deductions	REAL	Total deductions
netSalary	REAL	Net salary
workingDays	INTEGER	Working days in month
daysPresent	INTEGER	Days present
daysAbsent	INTEGER	Days absent
leaveDays	INTEGER	Paid leave days
status	TEXT	pending/paid
paidDate	DATETIME	Payment date
paymentMode	TEXT	cash/bank_transfer/cheque
processedBy	INTEGER	FK to Users
createdAt	DATETIME	Creation time
Unique: (staffId, month)		

Chapter 9

System Tables

9.1 ActivityLogs

Audit trail for all actions.

Column	Type	Description
id	INTEGER	Primary key
userId	INTEGER	FK to Users (nullable)
action	TEXT	login/logout/create/update/delete
module	TEXT	Module name
entityType	TEXT	Entity type (student, etc.)
entityId	INTEGER	Affected entity ID
description	TEXT	Action description
details	TEXT	Additional details (JSON)
previousValues	TEXT	Previous values (JSON)
newValues	TEXT	New values (JSON)
createdAt	DATETIME	Timestamp

9.2 Backups

Backup metadata.

Column	Type	Description
id	INTEGER	Primary key
fileName	TEXT	Backup file name
filePath	TEXT	Full file path
fileSize	INTEGER	Size in bytes
type	TEXT	manual/auto
description	TEXT	Backup notes
dbVersion	INTEGER	Database version
appVersion	TEXT	App version
studentCount	INTEGER	Students at backup time
staffCount	INTEGER	Staff at backup time
createdBy	INTEGER	FK to Users
isValid	BOOLEAN	Backup verified

Column	Type	Description
createdAt	DATETIME	Backup timestamp

9.3 NumberSequences

Sequential number generation.

Column	Type	Description
id	INTEGER	Primary key
name	TEXT	Sequence name (unique)
prefix	TEXT	Number prefix
suffix	TEXT	Number suffix
currentNumber	INTEGER	Current sequence value
minDigits	INTEGER	Minimum digits (pad zeros)
resetPeriod	TEXT	never/yearly/monthly
lastResetDate	DATETIME	Last reset date
includeYear	BOOLEAN	Include year in number
includeMonth	BOOLEAN	Include month in number
createdAt	DATETIME	Creation time
updatedAt	DATETIME	Update time